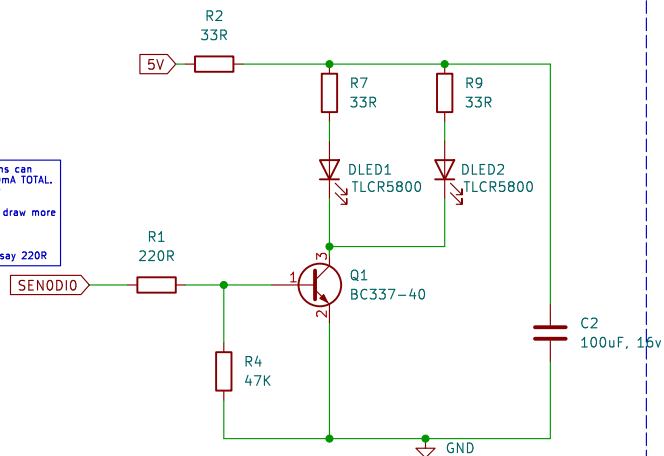
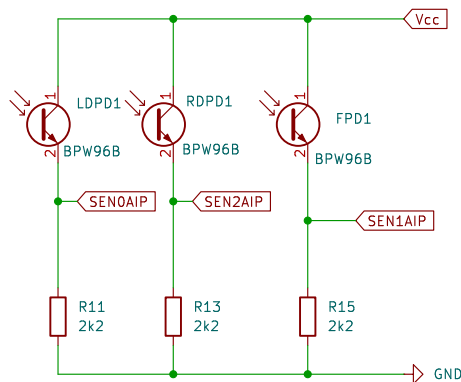


Diagonal Circuits

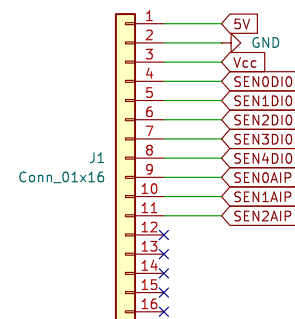
The RP2040 GPIO pins can support a max of 50mA TOTAL. RP2040 runs at 3.3v
Each pin should not draw more than 16mA
 $3.3/0.016 = 206$
Let's go higher and say 220R



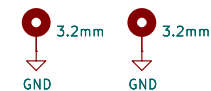
Detectors



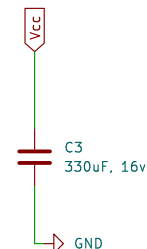
Board Connector to Mezzanine



Mounting Holes

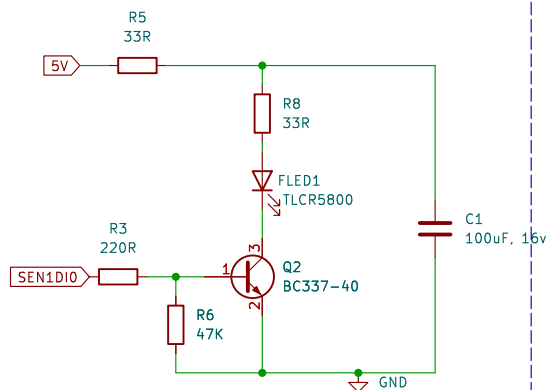


Power Smoothing

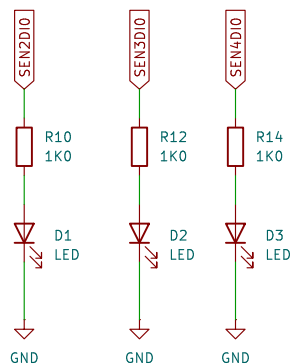


Front Circuit

The RP2040 GPIO pins can support a max of 50mA TOTAL. RP2040 runs at 3.3v
Each pin should not draw more than 16mA
 $3.3/0.016 = 206$
Let's go higher and say 220R



Indicator LED's



NOTES:

- SEN0DIO – "Diag LED" on PCB. Set High to turn on the diagonal sensor LED's
- SEN1DIO – "Frnt LED" on PCB. Set High to turn on the front sensor LED
- SEN2DIO – "Left LED" on PCB. Set High to turn on the left Indicator LED. D1
- SEN3DIO – "Mid LED" on PCB. Set High to turn on the middle Indicator LED. D2
- SEN4DIO – "Right LED" on PCB. Set High to turn on the right Indicator LED. D3
- SEN0AIP – "Left ADC" on PCB. Read via ADC pin to get value from the left photodiode. LDPD1
- SEN1AIP – "Front ADC" on PCB. Read via ADC pin to get value from the front photodiode. FPD1
- SEN2AIP – "Right ADC" on PCB. Read via ADC pin to get value from the left photodiode. RDPD1

Redesign of the UKMARS Advanced board to use SMD and Duncan's design

Sheet: /
File: UKMARSBOT_RP2040Wall-Sensor-Basic.kicad_sch

Title: Mouse SMD UKMARS Advanced Sensor Board

Size: A4
KiCad E.D.A. 8.0.9

Date: 2024-12-19

Rev: 1
Id: 1/1